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Survey of fishes, semi-aquatic turtles, and crayfishes on the Emporia State University Natural Areas

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Emporia State University (ESU) manages eight natural areas that offer a range of habitats, both aquatic and terrestrial, and are home to a variety of flora and fauna. Throughout the Flint Hills, some fish, turtle, and crayfish species have experienced declines in range and population. To track possible changes in assemblage composition with respect to these animal groups, it is vital to have a baseline understanding of species presence on these areas. The objectives of our study were to compile and organize historical data pertaining to fish, semi-aquatic turtle, and crayfish species' presence on ESU's Natural Areas and to complete a survey of the areas to create an accurate list of the species present on each site. Historical data consisted of field reports from ESU faculty members sent to the Director of the Natural Areas from 1997–present, past master's students' theses, current studies being performed, and communications with faculty members. Surveys were conducted from spring 2018–spring 2019. We found 42 fish, four semi-aquatic turtle, and two crayfish species on the ESU Natural Areas, including new species records on six of the eight areas. Our study created baseline species presence data for the ESU Natural Areas and may serve as a reference for future sampling on these sites.

Keywords: Habitat, populations, sampling, assemblages, species, baseline

Introduction

Emporia State University (ESU) manages eight natural areas: Campus Woods, Hamilton Fossil Quarry, Charles P. Coughlen Natural Area, F. B. and Rena G. Ross Natural History Reservation, Reading Woods Natural Area, Neva Marsh, Dunlap Bottoms, and Sarah Howe Natural Area (Jensen, Schulmeister, and Koerner 2016). The Emporia State Natural Areas lie in Lyon, Chase, Morris, and Greenwood counties on the eastern border of the Flint Hills in east-central Kansas. This area is primarily composed of tallgrass prairie and is one of the most endangered prairies on the planet (Samson and Knoft 1994). This region is now heavily affected by agriculture and urbanization, making water bodies especially endangered because what is left of the prairie is not large enough to create a natural watershed in most cases (Dodds et al. 2004).

Freshwater fish populations have experienced major declines within east-central Kansas (Haslouer et al. 2005; Perkin et al. 2015). Kansas lists 11 species of fish as threatened, and ESU's Natural Areas lie within the range of three of these species: Hornyhead Chub (*Nocomis biguttatus*), Topeka Shiner (*Notropis topeka*), and Neosho Madtom (*Noturus placidus*) (Anonymous 2019a). Kansas also lists 31 fish species as "species in need of conservation," and ESU's Natural Areas lie within the range of two of these species: Cardinal Shiner (*Luxilus cardinalis*) and Spotted Sucker (*Minytrema melanops*) (Anonymous 2019b). In recent decades, Flint Hills turtle populations have been changing (Busby and Parmelee 1995). Kansas lists one turtle species as threatened, the Northern Map Turtle (*Graptemys geographica*) (Anonymous 2019a), which can be found in Osage County directly to the east of Lyon County (Anonymous 2019c). Crayfish populations

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have also been diminishing, with 32% of North American crayfish species considered endangered or vulnerable (Glon et al. 2018). In addition to habitat change, invasive species have played a role in the declines of crayfish populations in the central United States. The Northern Crayfish (*Faxonius virilis*) is threatened by the introduction of the Rusty Crayfish (*Faxonius rusticus*) in North Dakota (Hayer et al. 2011). According to the Kansas Department of Wildlife, Parks, and Tourism (KDWPT), the Rusty Crayfish has not been found in Kansas, however this species is found in a neighboring state, Missouri, and, if introduced to Kansas, could threaten native crayfish populations (Anonymous 2019d). The Northern Crayfish is located across Kansas in permanent bodies of water (Ghedotti 1998).

Recent research pertaining to the ESU Natural Areas includes the rediscovery of Barred Tiger Salamanders in Lyon County (Sievert and Coleman 2013), inter-pond movements of Western Painted Turtles (House, Nall, and Thomas 2010), and habitat use by Neosho River fishes (Tiemann et al. 2004; Tiemann et al. 2005; Gillette et al. 2006). Due to the lack of a comprehensive list of fishes, semi-aquatic turtles, and crayfishes at each of ESU's Natural Areas, we sought to summarize all available historical data and conduct surveys to create a list of species occurring at each area. Historical data consisted of field reports and observation from ESU faculty members sent to the Director of the Natural Areas (Dr. W. Jensen) from 1997–present, past master's students' theses, current studies being performed, and personal communications with faculty members. Our objective was to provide a comprehensive list of the species of fishes, turtles, and crayfishes at all eight of ESU's Natural Areas.

Methods

Study Areas: Detailed descriptions of these areas were made by Jensen, Schulmeister, and Koerner (2016). Campus Woods is a 5-ha tract of land located directly north of the Emporia State University campus in Lyon County. It is dominated by riparian forest along the Neosho River. The Neosho River along this stretch is primarily pool habitat, with a few riffles interspersed. The Hamilton Fossil Quarry is a 21-ha tract located in Greenwood County, 5 km east of Hamilton. This land contains previously-used commercial rock quarries which have now been flooded. A small stream also runs north to south through the property. The Charles P. Coughlen Natural Area is an 18-ha tract 14.5 km southwest of Emporia in Lyon County. This area contains an unnamed tributary of Jacob Creek and a small impoundment, along with some riparian woodland (Jensen, Schulmeister, and Koerner 2016). The F.B. and Rena G. Ross Natural History Reservation is an 81-ha tract of land 24 km northwest of the Emporia State University's campus in Lyon County. The Ross Natural History Reservation primarily consists of tallgrass prairie, shrubland, and woodland. This reservation contains a small, unnamed stream, springs, and ponds, the largest of which is Gladfelter Pond (Fig. 1). Reading Woods Natural Area is a 15-ha tract of deciduous forest 24 km northeast of Emporia in Lyon County. Duck Creek, a tributary of the Marais Des Cygnes River, flows along its northern and eastern borders.

Neva Marsh is a 21-ha tract in the Cottonwood River floodplain in Chase County (Jensen, Schulmeister, and Koerner 2016). Restored to a wetland in 2005, this area now contains small impoundments that support aquatic vegetation and wildlife. Dunlap Bottoms is a 50-ha restored wetland in southeastern Morris County. This area was restored from cropland to a wet meadow in 2005. The Sarah Howe Natural Area, also known as Howe Woods, is a 5-ha tract of forest in eastern Lyon County. An ephemeral tributary of Duck Creek flows through this area.

Methodology: Fish surveys were conducted using a variety of aquatic sampling methods, including seining with a 4.5 m x 1.8 m x 4.8-mm-mesh seine, backpack electrofishing with a Smith-Root model 12 electrofisher

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Figure 1. Gladfelter Pond at the F.B. and Rena G. Ross Natural History Reservation. Photograph taken by Jackson Holmgren on 5 April 2019.

(Smith-Root, Vancouver, WA), cast netting with a 1.8 m radius x 9.5-mm-mesh net, and minnow traps baited with bread, 79 cm x 23 cm x 6.4-mm steel mesh with 5.7-cm openings at both ends (Tackle Factory, Fillmore, NY). Sampling methods varied depending on techniques appropriate for the given habitat, and was conducted until all habitat types had been sampled. Each natural area was surveyed for fishes once between March 2018 and May 2018, and once more between September 2018 and November 2018. Fishes were identified to species using Kansas Fishes Committee (2014).

Semi-aquatic turtles were surveyed utilizing baited, three-ring hoop traps, 1.83 m x 0.76 m x 2.54-cm mesh, (Memphis Net and Twine, Memphis, TN) and by hand capture. Hoop traps were set along shores of bodies of water on four of the Natural Areas: Campus Woods, Hamilton Fossil Quarry, Coughlen Natural Area, and the Ross Reservation. Traps were not set at the other four sites due to the lack of water between March 2018 and October 2018. Three hoop traps were placed at each site and baited with chicken liver placed in a PVC bait canister alongside punctured cans of cream corn. Once baited, traps were set at the site for a week, checked and rebaited every other day. Each site was sampled for two consecutive weeks between March 2018 and October 2018. Turtles were identified using Collins, Collins, and Taggart (2010).

Crayfish surveys were conducted on all eight Natural Areas in both aquatic and terrestrial habitats. Aquatic crayfish were collected by

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seine, 4.5 m x 1.8 m x 4.8-mm-mesh, and minnow traps baited with dog food (Thorp and Covich 2006), 79 cm x 23 cm x 6.4-mm steel mesh with 5.7-cm openings at both ends (Tackle Factory, Fillmore, NY).

Every trap set was collected the next day. Also, any crayfish seen were captured by hand or dip net. Williams and Leonard (1952) found collecting aquatic crayfish by seine to be the most efficient method of collection. Every pond, stream, and wadeable body of water was sampled at each site. Terrestrial crayfish were collected by excavating them from their burrows (Ridge et al. 2008). Collected crayfish were preserved in 10% formalin solution; after three days, specimens were rinsed in water and transferred to 40% isopropyl alcohol for identification (Pflieger 1996). Specimens were then identified to species using the dichotomous key in Pflieger (1996) and Crandall and De Grave (2017). Sampling was conducted between September 2018 and April 2019.

Historical records gathered from the Director of the Natural Areas consisted of personal communications from faculty members to the Director of the Natural Areas and annual field reports filed by ESU faculty members. The reports spanned from 1997–present. Records were also gathered from previous publications (Tiemann et al. 2004; House, Nall, and Thomas 2011). Current studies being conducted by graduate students Sam Schneider, Justin Autz, Jennifer Buchanan, and Michael Mahr were also used to supplement historic data. Results from past theses were also included in historic data (Nall 2009; Becker 2016).

Results

Campus Woods: Campus Woods had the most fish species of any of ESU's Natural Areas with 38 (Table 1). Noteworthy fish species in the area included the federally listed threatened species, the Neosho Madtom. No fish species not present in historical records were recorded for this area. Two semi-aquatic turtle species were collected at Campus Woods, the Red-eared Slider (*Trachemys scripta elegans*) and the Eastern Spiny Softshell Turtle (*Apalone spinifera spinifera*) (Table 1). Previously-recorded species absent from our collections were the Snapping Turtle (*Chelydra serpentina*), Ouachita Map Turtle (*Graptemys ouachitensis*), Mississippi Map Turtle (*Graptemys pseudogeographica kohnii*), and Midland Smooth Softshell Turtle (*Apalone mutica mutica*). Only one crayfish species was found at Campus Woods, the Northern Crayfish (Fig. 2; Table 1). This species had been recorded previously at this area.



Figure 2. Northern Crayfish captured during sampling. Photograph taken by Jackson Holmgren on 13 December 2018.

Hamilton Fossil Quarry: Fifteen fish species were found at the Hamilton Fossil Quarry, including two previously not known to inhabit this area, Largemouth Bass (*Micropterus salmoides*) and Spotted Sucker (Table 1). One semi-aquatic turtle species was found at this area, the Snapping Turtle, and was the only turtle species previously recorded (Table 1). One species of crayfish was found at this area, the Northern Crayfish (Table 1). No previous records concerning crayfish exist for this Natural Area.

Charles P. Coughlen Natural Area: A total of 16 species of fishes were collected at Coughlen Natural Area (Table 1). Previous records indicate the presence of the Sand Shiner (*Notropis stramineus*), Creek Chub (*Semotilus atromaculatus*), and Logperch

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Table 1. Historical records and fishes, semi-aquatic turtles, and crayfishes collected at ESU's Natural Areas during contemporary sampling from March 2018–November 2018

	Common Name	Scientific Name	Campus Woods		Hamilton Quarry		Coughlen Area		Ross Reservation		Reading Woods		Neva Marsh		Dunlap Bottoms		Sarah Howe	
			Historic	Current	Historic	Current	Historic	Current	Historic	Current	Historic	Current	Historic	Current	Historic	Current	Historic	Current
Fishes																		
	Longnose Gar	*Lepisosteus osseus*	X	X														
	Shortnose Gar	*Lepisosteus platostomus*	X															
	Gizzard Shad	*Dorosoma cepedianum*	X															
	Central Stoneroller	*Campostoma anomalum*	X	X	X	X	X	X										
	Bluntface Shiner	*Cyprinella camura*	X	X														
	Red Shiner	*Cyprinella lutrensis*	X	X	X	X						X						
	Common Carp	*Cyprinus carpio*	X	X														
	Redfin Shiner	*Lythrurus umbratilis*	X	X	X		X	X										
	Golden Shiner	*Notemigonus crysoleucas*	X															
	Ghost Shiner	*Notropis buchanani*	X	X														
	Carmine Shiner	*Notropis percobromus*	X	X														
	Sand Shiner	*Notropis stramineus*	X	X			X											
	Mimic Shiner	*Notropis volucellus*	X	X			X											
	Suckermouth Minnow	*Phenacobius mirabilis*	X	X														
	Bluntnose Minnow	*Pimephales notatus*	X	X	X	X	X	X				X						
	Slim Minnow	*Pimephales tenellus*	X															
	Bullhead Minnow	*Pimephales vigilax*	X	X														
	Creek Chub	*Semotilus atromaculatus*					X		X									
	Spotted Sucker	*Minytrema melanops*				X	X	X										
	Golden Redhorse	*Moxostoma erythrurum*	X	X	X													
	Black Bullhead	*Ameiurus melas*			X	X	X											
	Yellow Bullhead	*Ameiurus natalis*	X				X	X	X									
	Channel Catfish	*Ictalurus punctulatus*	X															
	Stonecat	*Noturus flavus*	X	X														
	Freckled Madtom	*Noturus nocturnus*	X															
	Neosho Madtom	*Noturus placidus*	X	X														
	Flathead Catfish	*Pylodictis olivaris*	X	X														
	Brook Silverside	*Labidesthes sicculus*	X		X								X					
	Blackstripe Topminnow	*Fundulus notatus*					X	X										
	Western Mosquitofish	*Gambusia affinis*	X	X	X	X	X	X				X	X		X			
	Green Sunfish	*Lepomis cyanellus*	X	X	X	X	X	X	X	X			X		X			
	Orangespotted Sunfish	*Lepomis humilus*	X		X	X	X	X										
	Bluegill	*Lepomis macrochirus*	X		X	X	X	X	X	X					X			
	Longear Sunfish	*Lepomis megalotis*	X				X	X										
	Spotted Bass	*Micropterus punctulatus*	X		X	X												
	Largemouth Bass	*Micropterus salmoides*	X			X	X	X	X	X								
	White Crappie	*Pomoxis annularis*	X															
	Orangethroat Darter	*Etheostoma spectabile*	X	X	X	X	X	X	X									
	Logperch	*Percina caprodes*	X	X	X	X	X					X						
	Channel Darter	*Percina copelandi*	X	X														
	Slenderhead Darter	*Percina phoxocephala*	X	X														
	Freshwater Drum	*Aplodinotus grunniens*	X															
Turtles																		
	Snapping Turtle	*Chelydra serpentina*	X		X	X	X	X		X								
	Western Painted Turtle	*Chrysemys picta belli*						X	X									
	Ouachita Map Turtle	*Graptemys ouachitensis*	X															
	Mississippi Map Turtle	*Graptemys pseudogeographica kohnii*	X															
	Red-eared Slider	*Trachemys scripta elegans*	X	X			X	X	X				X					
	Midland Smooth Softshell Turtle	*Apalone mutica mutica*	X															
	Eastern Spiny Softshell Turtle	*Apalone spinifera spinifera*	X	X														
Crayfishes																		
	Northern Crayfish	*Faxonius virilis*	X	X		X	X	X				X						
	Grassland Crayfish	*Procambarus gracilis*												X		X		

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(*Percina caprodes*), but these species were not collected in our survey (Table 1). Three semi-aquatic turtle species were found at this area: Red-eared Slider, Western Painted Turtle (*Chrysemys picta belli*), and Snapping Turtle (Table 1). Western Painted Turtles had not previously been recorded in the area. One crayfish species was found at this Natural Area: the Northern Crayfish (Table 1), matching previous field reports stating the presence of the species at this area.

F.B. and Rena G. Ross Natural History Reservation: Despite being the largest Natural Area, only three fish species were found (Table 1). This is likely due to all the stream habitat on the area being dry throughout the duration of our study, thus only Gladfelter Pond was sampled. The only semi-aquatic turtle species observed at this area was the Snapping Turtle, though Western Painted Turtles and Red-eared Sliders also occurred historically (Table 1). One crayfish burrow was found in April 2019, however no crayfish could be excavated (Table 1). Historical records noting crayfish for this area did not distinguish the species.

Reading Woods Natural Area: Despite its close proximity to major bodies of water, only four fish species were found at this area. All species discovered were new records for the area, as no fish sampling had been conducted previously at Reading Woods (Table 1). No semi-aquatic turtles were collected during this survey and no historical data exists for turtles at this area (Table 1). The Northern Crayfish was the only crayfish collected from this area, and no previous records exist concerning crayfish (Table 1).

Neva Marsh: This site was not sampled for fishes or semi-aquatic turtles because between March 2018 and November 2018 the area contained no water. However, in previous years, Green Sunfish (*Lepomis cyanellus*) and Western Mosquitofish (*Gambusia affinis*) were recorded in this area (Table 1). No historical data exists for semi-aquatic turtles here (Table 1). The Grassland Crayfish (*Procambarus gracilis*) was the only crayfish found (Table 1). Previous records for this area indicate the presence of crayfish, however did not distinguish species.

Dunlap Bottoms: Like Neva Marsh, fishes and semi-aquatic turtles were not sampled at this area because between March 2018 and November 2018 this area contained no water. However, in previous years Bluegill (*Lepomis macrochirus*), Green Sunfish, and Western Mosquitofish had been recorded here (Table 1). No historical data exists for semi-aquatic turtles at this site (Table 1). The Grassland Crayfish was the only crayfish found here and, no historical data for crayfishes exists for this area (Table 1).

Sarah Howe Natural Area: This area was not sampled for fishes or semi-aquatic turtles because between March 2018 and November 2018 it did not contain any water. No historical data for fish, turtles, or crayfish exist here.

Discussion

Historical data pertaining to fish, semi-aquatic turtle, and crayfish collections on the ESU Natural Areas is sparse and primarily confined to the Neosho River at Campus Woods, the Coughlen Area, and the Ross Reservation. Sites at these areas have been visited repeatedly for class field trips since the 1950s, and sites within Campus Woods have been studied by master's students from ESU, leading to an extensive list of fish species being available for these areas. Apart from Campus Woods, the Coughlen Area, and the Ross Reservation, data pertaining to the Natural Areas is primarily found only within a few annual reports produced by ESU faculty members regarding their activities conducted on the areas (primarily class field trips). Although voucher species were not available for most of these reports, records in this study were verified by ESU faculty members.

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Given the aforementioned data available for Campus Woods, the Coughlen Area, and the Ross Reservation it was not surprising that no new species were collected on either of those areas. However, with the presence of John Redmond Reservoir (a large, man-made reservoir stocked for sportfishing) 25 km downstream from Campus Woods, it is conceivable that sportfishes stocked in the lake, such as Wiper (*Morone saxatilis x M. chrysops*), Blue Catfish (*Ictalurus furcatus*), and Paddlefish (*Polyodon spathula*) could occur on the area in the future. Although only 24 of the 38 species known to occur within Campus Woods were collected during our study, there is no reason to believe that some species have been extirpated, as the habitat within Campus Woods is large, and a large-scale project would be needed to comprehensively sample the fish community of the area.

Within the Coughlen Area, Sand Shiner, Creek Chub, and Logperch were absent from collections during our study, but had been present historically. Absence of these fishes from the Coughlen Area could have been due to their inability to repatriate the stream reach after desiccation, given the presence of multiple beaver dams on the area and a man-made dam downstream. If these species had been extirpated from the area during a period of drought, repatriation from downstream may have been blocked.

All species collected within Reading Woods were new records for the area, as no historic data existed pertaining to the fish community existed here due to a lack of previous sampling. The stream that flows through the area is intermittent. Given the close proximity of the stream reach to the Marais des Cygnes River and Melvern Reservoir, it is likely that other species will occur on the area during future periods of higher flow.

Perhaps the Natural Area with the most potential for new records of fish species is the Hamilton Fossil Quarry. This is due to a lack of verified historic data and only two fish collections being conducted in the stream and ponds on this area.

Neva Marsh, Dunlap Bottoms, and Sarah Howe Woods were dry from fall 2018 to spring 2019, but, baseline data pertaining to the fish communities of Neva Marsh and Dunlap Bottoms exists in the form of historical records taken from reports detailing class field trips to the areas. At the Ross Natural History Reservation, the only water present was Gladfelter Pond, which explains why stream-dwelling species such as Creek Chub and Orangethroat Darter (*Etheostoma spectabile)* were not collected during our study.

At Campus Woods, four semi-aquatic turtle species that had been historically found were not observed during this survey. While possible, it is not likely that these species are now absent from this area, as none is listed as threatened or endangered, and Campus Woods is well within their range. It is more likely that the species were simply not caught, potentially due to flooding, trap locations, escape, or the relatively small sampling effort. That Red-eared Sliders were not found at the Ross Natural History Reservation is likely due to similar causes, as Gladfelter Pond was surveyed later in the season, and after high amounts of rain. This species was also historically found at Neva Marsh, but not recorded in our survey because Neva Marsh was not sampled due to the area not containing any water from spring 2018 to fall 2018. A new record for the Western Painted Turtle was recorded at the Coughlen Natural Area.

The only historical records that exist for crayfishes on the ESU Natural Areas are field notes and observations made by faculty members at ESU. For Campus Woods and the Coughlen Area, these records state the presence of the Northern Crayfish. Other field notes exist for Neva Marsh, however only state the presence of crayfish and do not identify the species. The Northern Crayfish has been

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found in the drainage ditch immediately west of Dunlap Bottoms; thus, it is conceivable that during high water events this species could move into the area. It was not a surprise to find Northern and Grassland crayfish at ESU's Natural Areas because these areas lie within the range of these species. Crayfish burrows were found at Campus Woods and the Ross Natural History Reservation, however no crayfish were collected from them. This suggests the presence of a burrowing crayfish species at these areas, most likely the Grassland Crayfish.

We recommend that future surveys be conducted in order to build upon the species lists created by our study, as well as to monitor for changes in species presence on the natural areas. Also, further study of the fishes, turtles, and crayfishes of Neva Marsh, Dunlap Bottoms, and Sarah Howe Woods during periods of adequate water would be beneficial, as these areas were dry during the time period of our study.

The Emporia State University Natural Areas provide diverse habitats for fishes, semi-aquatic turtles, and crayfishes. These areas are a great asset for Emporia State students and faculty alike. Our list of fishes, semi-aquatic turtles, and crayfishes found at these areas help establish a baseline for future surveys of the aquatic and semi-aquatic fauna at the ESU Natural Areas. We recommend an increased sampling effort for future surveys at the ESU Natural Areas.

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